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Machine Learning Based Automated Industrial-Grade Hospital Waste Segregation Cyber-Physical System for Sustainable Healthcare

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Abstract

The improper management of biomedical waste in healthcare organizations represents escalating environmental, occupational, and operational challenges. More than 5.9 million tonnes of medical waste are generated annually all over the world, and approximately 15 percent of this waste is classified as hazardous. This poses serious health hazards, infection risks, and high treatment costs. The study proposes a machine-learning based cyber-physical framework that enables smart hospital waste segregation, elevates worker safety, and enhances sustainable healthcare operations with resource efficiency. A two-stage deep learning pipeline integration of YOLOv8, EfficientNet-B0, and ResNet-50 accurately identifies hazardous, non-hazardous, biodegradable and non-biodegradable waste streams. The architecture is an edge AI system that is tested and validated at the software stage and is driven by the need to provide energy efficient inference and real-time decision making, hence making it applicable to the industrial deployment. The outcomes highlight great potential to decrease human exposure to infectious materials, decrease landfill and incineration loads and optimize waste lifecycle performance and lead to circular, resource-efficient, and environmentally sustainable healthcare waste management systems.

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1. Introduction

Biomedical waste has been steadily increasing in the healthcare facilities, and it has become a major environmental and occupational hazard. According to World Health Organization (WHO), approximately 85 percent of health care waste falls under non-hazardous category; and the other 15 percent under hazardous category which comprises of infectious, toxic or radioactive waste that contribute greatly to

global morbidity and mortality [1]. Hospitals in low-income areas generate 0.2-0.5 kg of hazardous waste per bed in a single day, and the amounts of waste has drastically increased during the COVID-19 pandemic [1]. Nearly a third of healthcare facilities do not have safe disposal methods and contribute to a significant proportion of the 5.2 million deaths (approximately) occurring annually due to improper management of biomedical waste, out of which about 4 million are associated with children. This is attributable to the

poor management of medical waste [2]. Manual waste handling exposes more than two million healthcare workers to infectious agents that lead to the contraction of hepatitis B, C and, HIV [1,2]. This clearly highlights the necessity for implementing sustainable and systemic interventions in current frameworks to ensure protection of human health, reduce emissions, and address the degradation of environment throughout the waste management lifecycle [23].

Traditional biomedical waste disposal mechanisms are mainly manual and semi-automated, with fundamental sorting which is ineffective, inconsistent, and unsafe [3,4]. Even conveyor or sensor-based automation systems still need human monitoring and cannot eliminate public exposure to infectious wastes and harmful processes [5,6]. With the increasing volumes of waste around the globe, these existing systems cannot sustain current and future sustainability and safety demands. This highlights the necessity to move towards AI-based lifecycle-efficient cyber-physical systems (CPS) which will help decrease the intensity of energy consumption, enhance material recovery, and reduce occupational hazards. [7,8,9].

A multi-stage, robotic arm-powered conveyor sorting system, presented in this study, addresses the problems through machine learning, deep vision models, and robotic automation to provide sustainable waste management. The innovation of the system lies in the chain of multiple layers of automation that combines Autonomous Path Vehicles (APVs) to collect and transfer waste, a two-stage robotic sorting station and a control layer is used in the synchronization and functioning of the automation system. APVs follow the predetermined service routes to safely transport waste in hospital areas to segregation block and reduce the amount of human exposure incurred in the collection process. In these units, a two-stage classification pipeline is run in real-time: YOLOv8 is used to detect and localize objects [11,12,19], EfficientNet-B0 is tasked to classify those detected wastes as hazardous or non-hazardous, and ResNet-50 will further classify the waste into biodegradable and non-biodegradable materials [15,16,17]. Sharps, residual chemicals, infectious wastes among other hazardous materials are disposed to secure modules of disposal [13,14]. Recyclable, and compostable wastes, are redirected into recovery processes. Such AI-based vision systems and robotic arms, in coordination with CPS communication helps to reduce human contact, infection risk and as well as increase efficiency in the workflow [18,19].

The suggested CPS structure has been tested and validated at the software level and has the potential to create a platform of environment friendly and circular healthcare processes. By creating extremely specific classification, the system diverts the load from causing landfill and incineration by the efficient separation of recyclable and biodegradable fractions to the corresponding recovery streams [20,23]. It reduces power usage and carbon emissions by deployment of energy efficient edge AI processing units [21] which improves environmental performance. Moreover, the proper classification will help lead to the reuse of the materials, resulting in metals, plastics, and organic fragments returning to the cycles of productivity, thus enhancing resource efficiency, promoting circularity in

waste management, and reducing material losses (UN SDG 12) [6,20]. Autonomous robotic handling paired with predictive control optimizes throughput, minimizes maintenance cycles, and drastically improves lifecycle cost efficiency. Above all, the automation at the stages of collection, sorting, and disposal can reduce the exposure of workers to infectious substances, which strengthens occupational safety and environmental protection [22]. In general, this framework encourages the closed-loop, low-emission, and resource-optimal management of biomedical waste that assists in supporting sustainable healthcare facilities, long-term operational sustainability, and the ongoing enhancement of the environmental performance.

2. Background and Related Work

2.1. Existing Waste Segregation Systems

The three major types of biomedical waste segregation include manual, semi-automated, and automated systems. In low-resource settings (hospitals), manual segregation is a common practice. It is labour-dependent, inconsistent, and it exposes workers to hazards [2,3]. Semi-automation with colour coded bins and conveyor sorting reduces some of the risks, but nonetheless exposes the staff to infectious material [5]. Fully-automated systems are still confined to wet-dry or recyclable/non-recyclable separation [20,22]. Although there are a few advantages, these techniques have several shortcomings: manual sorting is ineffective; semi-automation is not scalable; automated sorters, are solely dependent upon weight, moisture or infra-red signal. These systems are rigid and can process only simple waste streams [11,13,16]. Additionally, they all are not well adapted to the dynamic aspect of the biomedical waste, with no immediate feedback and dynamic intelligence needed in the current healthcare [7].

2.2. Limitations of Existing Systems

Key constraints to the traditional systems of managing biomedical waste are as follows:

- Lifecycle risk and Safety deficiencies:

Globally, about 2 million healthcare workers handle biomedical waste manually, and face the risk of being exposed to infectious or toxic substances during collection. This results in occupational hazards as well as exposes workers to the risk of pathogen transmission during the waste management lifecycle [2, 3].

- High Carbon and Energy footprint:

The semi-automated segregation lines depend on continuous power supply and poor transport performance, which lead to high levels of emissions – consume up to 25-30 % more energy-per tonne of waste than what is required for localized systems [5, 8].

- Limited Material Recovery and Inefficiency in Resources:

Downstream recycling and recovery is severely limited due to the mixed disposal of plastics, metals, and organics with hazardous waste. The lack of digital traceability also reduces

lifecycle material flow analytics needed to optimize the use of resources in the circular system (UN SDG 12; [6, 20]).

- Excessive Disposal Burden:

Contamination and misclassification direct the waste fractions toward incineration or cause landfills which can increase greenhouse gas emissions and resource loss rather than using them to provide end-of-life pathways that promote sustainable disposal of the generated waste [6, 20,23].

2.3. Advances in AI and Cyber-Physical Systems

Recent years have become a time of great breakthroughs in the field of artificial intelligence (AI), computer vision, and deep learning models which are used actively in biomedical waste management today. Advanced networks like ResNet, EfficientNet, DenseNet and Vision Transformers have been shown to perform tasks linked with image-based classification with high accuracy (over 95% in medical image recognition benchmarks) [10,12,17]. Using transfer learning, such models can be generalized to obtain outstanding accuracy in classifying waste streams even in situations of contamination or low-light conditions [11,15,16]. Cyber-physical systems (CPS) combine sensory information (infrared sensor, RGB camera), actuator control (robotic arm, conveyor mechanism) and AI (intelligent decision-making) into a single platform [8,9]. Load based waste segregation paired with real-time response control helps in minimizing the risk of health hazards. Such systems make it possible to monitor waste in real-time, eliminating most of the human intervention involved [7,13,14]. IoT and cloud-based analytics integration assists in facilitating data-driven information to settle policy-making, forecasted maintenance and waste management advancements [6,13,20].

2.4. Research Gap and Motivation

Even after significant breakthroughs, the literature notes that most of the actual AI-based waste separating machines remain at the prototyping stage, or in the laboratory setting [7,15]. In recent publications, the emphasis has been placed more precisely on dry-recyclable vs. wet-organic, and less focus on hazardous biomedical waste segregation [3,4]. Further, research is minimal in integrating deep learning, robotics, and infrared sensing into a scalable and deployable CPS framework that can sort heterogeneous biomedical waste at hospital scale [8,9,14].

3. System Design and Methodology

The proposed technological solution is a cyber-physical waste sorting system that can be integrated into the existing production line of biomedical wastes and consists of computer vision models and embedded platforms, robotic actuation, and sensor-based validation to provide automated sorting and handling of biomedical or healthcare wastes. The method is designed as a two-layered system (i): the software intelligence layer, comprising convolutional neural networks (YOLOv8, EfficientNet-B0, ResNet-50), is used to recognize and classify waste [10,17], and (ii): the hardware execution layer, which

consists of NVIDIA Jetson modules, robotic arms, conveyor mechanisms, and microcontrollers handle real-time operation control and decision-making [18,21]. Fig.1 represents the general structure of the system where the combination of machine learning, sensor fusion, and robotic manipulation are integrated into a scalable system [7,14].

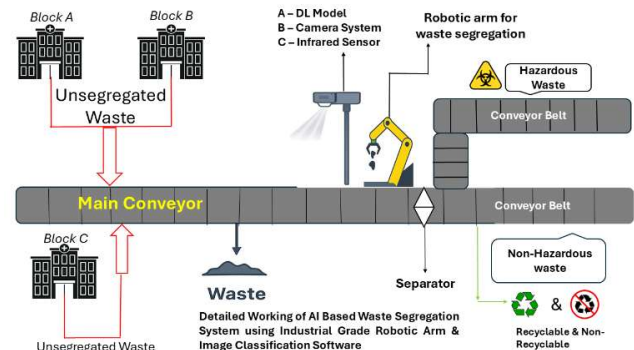


Fig. 1. Detailed illustration of hazardous and non-hazardous waste separation

3.1. Dataset Preparation and Preprocessing

The dataset includes 7,800 images that were gathered through various sources in order to be diverse and representative. Approximately 60 percent of samples were taken from open-source repositories (Kaggle, Roboflow), 25 percent were collected in-house, and 15 percent were taken out of licensed publicly reusable databases. The images were categorized into hazardous and non-hazardous with the latter further divided into biodegradable and non-biodegradable as per the biomedical-waste regulations [1, 3]. The image division was done according to 70:20:10 ratio for training, validation and testing respectively. Non-overlapping sets were checked through hash-based detection. Perceptual aHash was used with Hamming threshold <5 and cross-split duplicates were identified and removed. [15, 16]

The biodegradable and non-biodegradable samples included 5 subclasses, and were divided into 2,340 and 2,730 samples respectively. The hazardous type included 10 subclasses (273 base images each) which are extensively and precisely augmented to balance scarcity and to enhance the model generalization [7, 11]. Only the train images have been augmented to ensure that test images maintain authenticity and resemble real world conditions. The per-class distribution of images in the dataset is shown in Table 1.

Diverse lighting conditions, as well as background conditions, were used to obtain images and make them more realistic in terms of representativeness of hospital settings. A 24 MP APS-C DSLR camera with 50 mm f/1.8 lens and a combination of fluorescent and ambient light dispersed with reflectors was used to take in-house samples. The objects were put on neutral, matte or contextual hospital surfaces and had a representative visual diversity, with exposure and framing of an equal standard [7, 13, 14].

The images were resized to assist classification (224x224 px) and detection (640x640 px) tasks and balanced using normalization. Geometric (rotation, flipping, scaling, cropping) and photometric (brightness, contrast, hue, saturation, local contrast) transformations were integrated into

the pipeline of these models to improve the generalization of the models [10-12, 16]. Approximately, 15 percent of the training images were subjected to synthetic background substitution by textures of hospitals to simulate real world conditions of the conveyor-belts.

Roboflow has been used to generate and verify annotations, and 10% of the annotations were double- labelled (percent agreement=96%, mean pairwise IoU (or) Intersection over Union = 0.92). One-to-one image-annotation mapping and the removal of near-duplicates were done by automated quality filters (Hamming < 5). The metrics of the datasets comprised 3.8 ± 0.6 MP average resolution, 35 to 70% object-to-frame ratio, and sharpness > 0.85 (Laplacian variance). Gathering of the waste samples was done carefully and the imaging process was carried out responsibly without including any patient or individual data.

Table 1. Distribution of Images Across Classes

Category	Sub-class	Train	Val	Test	Total
Hazardous	10 subclasses	1,911	546	273	2,730
Non-Hazardous–Biodegradable	5 subclasses	1,638	468	234	2,340
Non-Hazardous–Non-Biodegradable	5 subclasses	1,911	546	273	2,730
Total	—	5,460	1,560	780	7,800

3.2. Deep Learning Model Development

- YOLOv8 – Object Detection and Localization:

The YOLOv8 model has been utilized to handle the unstructured intrusion of waste materials into conveyor belts by localization and real-time object detection. It operates on a C2f backbone paired with a PAN/FPN-style neck, which enables multi-scale feature fusion and the detection head is highly efficient in predicting bounding boxes and class probabilities [11,12,19]. They operate at very high accuracy using 640x640 input, and with inference speeds of 30-50 FPS on GPUs; therefore, can be easily integrated into robots.

- ResNet-50 for Biodegradable vs Non-Biodegradable (Binary) Classification:

The waste items classified as non-hazardous by YOLOv8 model are further split into biodegradable and non-biodegradable and this part of the classification is done by ResNet-50. Its residual architecture, enforced with 50-layers, is enhanced with skip connections, and routed to overcome vanishing gradient and outperform fine-grained texture classification [12]. This functionality is required in the separation of biodegradable wastes i.e. food, tissues and paper, and non-biodegradable plastics or packaging materials. ResNet-50 processes the images of a standard size (224 x 224 px) and depicts stable performance on the scaled-up datasets.

- EfficientNet-B0 for Hazardous vs. Non-Hazardous (Binary) Classification:

EfficientNet-B0 was incorporated to carry out the main hazardous vs. non-hazardous classification task. The depth, width and input-resolution balance concept of its compound scaling principle enables it to be highly accurate while reducing the computational costs. EfficientNet-B0 consists of only 5.3 million parameters and takes images of 224x224 input resolution, and therefore it is relatively easier to be deployed over embedded systems like the NVIDIA Jetson [21]. This architecture is especially immune to scale, light, and orientation effects, which are common in the natural contexts of hospital waste streams, and it is possible to precisely identify solid and liquid hazardous waste like syringes, scalpels, blood-stained objects and chemical containers. Fig. 2 illustrates the image processing and classification pipeline carried out by all three models.

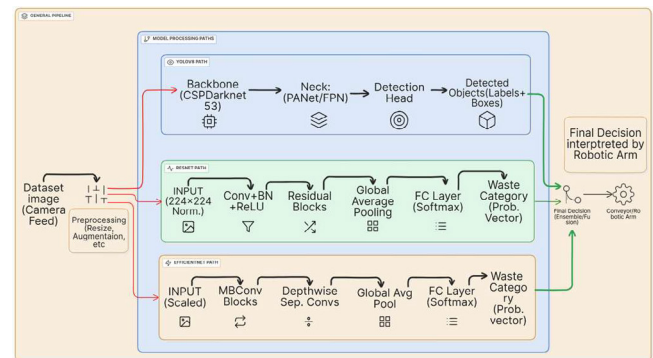


Fig. 2. Pipeline Based Illustration of Three Deep Learning Models

3.3. Model Training and Results

The training was implemented on a Dell G15 3350 workstation, which has an NVIDIA RTX 3050 GPU (6 GB VRAM) and 16GB of RAM. All the models were optimized with the Adam optimizer that was applied with a batch size of 32 and learning rate set to 0.001. This optimizer has been chosen due to its computational efficiency and the ability to control the learning rate per-parameter and more specifically it is optimally suited to large and complex models [12]. ResNet-50 and EfficientNet-B0 had 100 epochs, and YOLOv8 had 150 epochs so as to be trained in accordance with its multilayer complexity of localization and image detection. The loss-curve of training and validation loss values of each model were gradually descending as the epochs increase. The stable convergence of the loss curves highlight the model's balanced performance with no evident overfitting.

- YOLOv8 (Object Detection): The training loss reduced from about 2.09 at the first epoch to about 0.4 at the final epoch (total of 150 epochs). The validation loss decreased from 2.25 to about 0.51 at the last epoch.
- ResNet-50 (Biodegradable vs. Non-Biodegradable Classification): As per the training loss, it has come down from roughly 1.6 to 0.3 and the validation loss ranges from 1.75 to about 0.4 in a span of 100 epochs.

- EfficientNet-B0 (Hazardous vs. Non-Hazardous Classification): over 100 epochs, the training loss has reduced to 0.25 from an initial value of 1.34 and the validation loss reduces to 0.37 from 1.48 at the first epoch.

Fig. 3(a) and Fig. 3(b) illustrate the convergence of loss over number of epochs during Training and Validation respectively. YOLOv8 was trained on an additional 50 epochs compared to ResNet-50 and EfficientNet-B0 due to its complex task of localization on images and object detection properties [11,19]. Table 2. Represents the primary performance metrics of deep learning models.

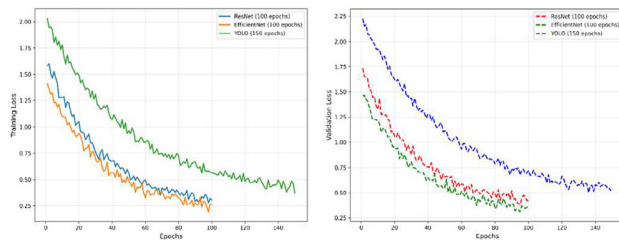


Fig. 3 (a) Training Loss vs Epochs (b) Validation Loss vs Epochs

Table 2. Standard Performance Metrics of All Three Models

Model	Task (Classification)	Primary Metric(%)	Precision (%)	Recall (%)	F1-Score (%)
YOLOv8	Object Detection and Localization	94.8 (mAP@0.5)	94.2	95.1	94.6
ResNet-50	Biodegradable vs. Non-Biodegradable	93.7 (Accuracy)	92.9	93.6	93.2
EfficientNet-B0	Hazardous vs. Non-Hazardous	97.2 (Accuracy)	95.6	96.3	96.1

In the case of YOLOv8, the mAP@0.5 (mean Average Precision of an IoU of 0.5) is considered as the key detection measure, which complies with the standard object detection protocols. Assessment of inter-class agreement beyond chance is done by calculating the Cohen's Kappa, and the result is $k = 0.89$ with EfficientNet-B0, and $k = 0.87$ with ResNet-50. The high predictive consistency of all types of waste is also confirmed further by the per-class evaluation results achieved through the confusion matrices as depicted in Fig.4, and through the computation of Matthews Correlation Coefficient (MCC) of both EfficientNet-B0 (0.88) and ResNet-50 (0.85).

Industrial grade RGB cameras and NVIDIA Jetson Xavier were employed to assess real-time image classification. The system achieved a throughput of 35-38 FPS, inference time of ~27ms (~37 FPS), and 63 percent of GPU utilization with a thermal stability of less than 65°C. On-device mAP@0.5 was over 94% and the end-to-end latency was around 65ms. This aligns with the aim of circular waste management processes, the system has the potential to reduce exposure to bloodborne pathogens by up to 60% [2], GHG emissions can be reduced by 70-88% through reduced incineration [1,3], landfill reliance reduction by about 30-50% and enable 40-60%

recovery of the 85% non-hazardous biomedical waste fraction reported by WHO [1,6,20].

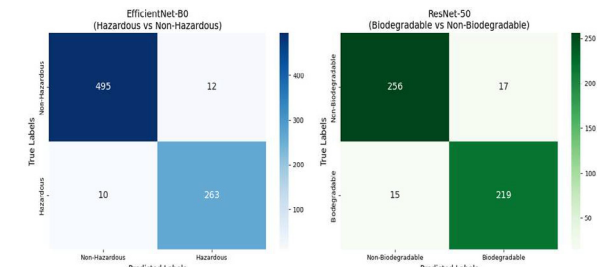


Fig. 4. Recall values above 90% were achieved on the test set through per-class evaluation for sub-class level detection, thus depict consistent results.

3.4. Cyber-Physical System Integration and Deployment

Cyber-physical architecture is the backbone of the entire automated waste-segregation system with NVIDIA Jetson being the core, high-performance edge AI processing unit. Such platform is precisely chosen based on its capability to close the integration gap between software intelligence (deep learning models) and hardware execution (robotics and conveyor systems).

NVIDIA Jetson (e.g., AGX Xavier) is used to connect the hardware components owing to its high computational speed in a small, power-constrained form factor. It can run extensive deep learning inference at edge with unprecedented capability, avoiding the network and processing delays and bandwidth bottlenecks that occur frequently due to data processing over the cloud [21]. Pretrained models including YOLOv8, ResNet-50, and EfficientNet-B0 can be run directly on the powerful Jetson GPU using optimized software frameworks that include PyTorch and TensorFlow. This enables this system to undertake real-time, on-device detection and classification of objects. The trained models can be implicitly transformed into a narrowly-optimized, serial form (i.e. ONNX) so as to maximize cross-platform compatibility and execution on the Jetson hardware [21].

TensorRT can be further optimized by using NVIDIA TensorRT to make the models optimized to run on the Jetson specific architecture. These operations may involve layer/kernel merge, precision calibration (e.g., INT8 quantization), and memory compactness, which may greatly maximize inference speed and instance unit power. Fig.5 illustrates the waste segregation pipeline and its working.

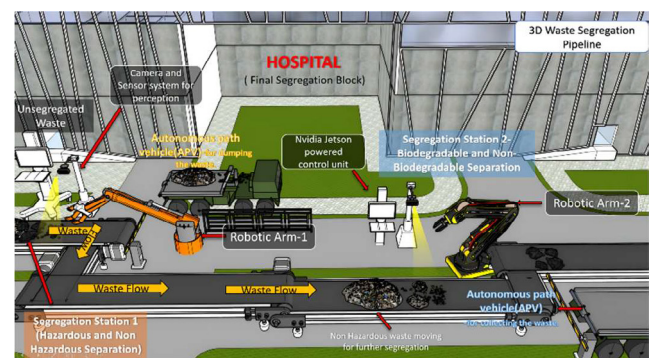


Fig. 5. 3D software stage illustration of the entire cyber-physical system

The central control unit is the NVIDIA Jetson, which handles real-time processing of high-resolution images from industrial grade RGB cameras (and optionally infrared sensors) [8,21]. This AI analysis facilitates the whole pipeline. The robotic arms include embedded microprocessors (i.e., ARM-based microcontrollers such as Arduino or ESP32 boards) and are linked to the Jetson by a powerful communications protocol such as Ethernet or high-speed serial interface. Depending on its real-time AI analysis, the Jetson is able to issue high-level commands such as MOVETO(x, y) or GRABITEM(class_ID). Those commands can be fed to the microprocessor of the robotic arm, which is then processed into exact motor control commands and sent to the servo motors driving the arm and the necessary physical motion can be performed. This distributed control architecture can offload low-level tasks off the Jetson to enable the system to work with coordinated precision and maintenance of minimal latency [9]. Similarly, the conveyor control signals like stopping or rolling the belt are also managed by Jetson depending upon the detection of waste materials.

4. Conclusion and Future Works

The proposed solution of AI-based biomedical waste segregation that uses EfficientNet-B0, ResNet-50, and YOLOv8 to build an effective cyber-physical framework is a feasible and safer waste management system for hospitals. The system has been tested and assessed at the software level and demonstrates great prospects of full-scale robotic implementation through high classification and operational efficiency while minimizing human exposure to infectious materials. The framework contributes to some of the most important sustainability results emphasized by the concepts of Life Cycle Engineering (LCE): minimizing landfills and incineration by means of accurate segregation and recycling, minimizing emissions and energy use by less carbon-intensive treatment techniques and improving health and safety of the workers through automation of processes. These outcomes correspond to the principles of circular economy and directly align with the WHO and relevant UN SDG, particularly SDG 3, SDG 11, and SDG 12, which focus on responsible consumption, safer working conditions, and sustainable healthcare practices. Further research efforts will be on pilot-scale deployments, larger datasets, regional analytics, IoT integration, and a screening life cycle assessment (LCA) to measure the prevention of incineration load (kg CO₂-eq/tonne) and energy intensity (kWh/tonne) compared to traditional manual or semi-automated segregation systems.

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